ISLANDS AND CORAL REEFS / INDIA ISLANDS-GREAT NICOBAR

ISLAND-ORIGIN TREE → REEF-BUILDING ENGINE → REEF-TYPE CROSS-SECTION → DARWIN SEQUENCE → DALY QUALIFICATION → BLEACHING OUTCOME FORK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ● SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

00

STAGE 00 ISLAND-ORIGIN TREE

ISLAND-ORIGIN TREE · CRUST OR SHELF AFFINITY · MAGMA BUILDS OCEANIC EDIFICE · REEF CARBONATE + WAVE-BUILT SEDIMENT · RIVER, CURRENT OR BARRIER ACCUMULATION · MUST REMEMBER · ISLAND ANALYSIS DISTINGUISHES CONTINENTAL, VOLCANIC, CORAL AND DEPOSITIONAL ORIGINS

CONTINENTAL → crust or shelf affinity

VOLCANIC → magma builds oceanic edifice → CORAL → reef carbonate + wave-built sediment → DEPOSITIONAL → river, current or barrier accumulation

CORE CLAIM
- CONTINENTAL → crust or shelf affinity
- VOLCANIC → magma builds oceanic edifice

MECHANISM
- CORAL → reef carbonate + wave-built sediment
- DEPOSITIONAL → river, current or barrier accumulation

CONSEQUENCE / LIMIT
- MUST REMEMBER: Island analysis distinguishes continental, volcanic, coral and depositional origins
- reef-building follows light, temperature, salinity, substrate and symbiosis, with fringing–barrier–atoll sequences compared against Lakshadweep and Andaman-Nicobar tectonic settings.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Island analysis distinguishes continental, volcanic, coral and depositional origins; reef-building follows light, temperature, salinity, substrate and symbiosis, with fringing–barrier–atoll sequences compared against Lakshadweep and Andaman-Nicobar tectonic settings.

01

STAGE 01 REEF-BUILDING ENGINE

REEF-BUILDING ENGINE · CALCIUM-CARBONATE SKELETON · PHOTOSYNTHETIC SYMBIOSIS · LIGHT + CLEAR WATER · REEF PRODUCTIVITY · SEDIMENT STRESS · BLEACHING OR DECLINE

POLYP → calcium-carbonate skeleton → ZOOXANTHELLAE → photosynthetic symbiosis → LIGHT + CLEAR WATER → reef productivity → HEAT / SEDIMENT STRESS → bleaching or decline

CORE CLAIM
- POLYP → calcium-carbonate skeleton
- ZOOXANTHELLAE → photosynthetic symbiosis

MECHANISM
- LIGHT + CLEAR WATER → reef productivity
- HEAT / SEDIMENT STRESS → bleaching or decline

CONSEQUENCE / LIMIT
- POLYP → calcium-carbonate skeleton
- ZOOXANTHELLAE → photosynthetic symbiosis

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HEAT / SEDIMENT STRESS → bleaching or decline.

02

STAGE 02 REEF-TYPE CROSS-SECTION

REEF-TYPE CROSS-SECTION · SHORE CONTACT + NARROW LAGOON · OFFSHORE REEF + BROAD LAGOON · RING REEF AROUND LAGOON · LAND RELATION, LAGOON AND FOUNDATION

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
FRINGING → shore contact + narrow lagoon	ATOLL → ring reef around lagoon	FRINGING → shore contact + narrow lagoon
BARRIER → offshore reef + broad lagoon	TEST → land relation, lagoon and foundation	BARRIER → offshore reef + broad lagoon

CORE CLAIM
- FRINGING → shore contact + narrow lagoon
- BARRIER → offshore reef + broad lagoon

MECHANISM
- ATOLL → ring reef around lagoon
- TEST → land relation, lagoon and foundation

CONSEQUENCE / LIMIT
- FRINGING → shore contact + narrow lagoon
- BARRIER → offshore reef + broad lagoon

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ATOLL → ring reef around lagoon; TEST → land relation, lagoon and foundation.

03

STAGE 03 DARWIN SEQUENCE

DARWIN SEQUENCE · VOLCANIC ISLAND + FRINGING REEF · SLOW SUBSIDENCE + UPWARD CORAL GROWTH · BARRIER REEF + WIDER LAGOON · ISLAND SUBMERGED

VOLCANIC ISLAND + FRINGING REEF → SLOW SUBSIDENCE + UPWARD CORAL GROWTH → BARRIER REEF + WIDER LAGOON → ISLAND SUBMERGED → ATOLL

CORE CLAIM
- VOLCANIC ISLAND + FRINGING REEF
- SLOW SUBSIDENCE + UPWARD CORAL GROWTH

MECHANISM
- BARRIER REEF + WIDER LAGOON
- ISLAND SUBMERGED → ATOLL

CONSEQUENCE / LIMIT
- VOLCANIC ISLAND + FRINGING REEF
- SLOW SUBSIDENCE + UPWARD CORAL GROWTH

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BARRIER REEF + WIDER LAGOON; ISLAND SUBMERGED → ATOLL.

04

STAGE 04 DALY QUALIFICATION

DALY QUALIFICATION · GLACIAL SEA-LEVEL FALL · EXPOSED PLATFORM · WAVE PLANATION · FOUNDATION SHAPING · POST-GLACIAL RISE · RENEWED REEF GROWTH · COMBINE SEA LEVEL, BIOLOGY AND TECTONICS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
GLACIAL SEA-LEVEL FALL → exposed platform	POST-GLACIAL RISE → renewed reef growth	GLACIAL SEA-LEVEL FALL → exposed platform
WAVE PLANATION → foundation shaping	VERDICT → combine sea level, biology and tectonics	WAVE PLANATION → foundation shaping

CORE CLAIM
- GLACIAL SEA-LEVEL FALL → exposed platform
- WAVE PLANATION → foundation shaping

MECHANISM
- POST-GLACIAL RISE → renewed reef growth
- VERDICT → combine sea level, biology and tectonics

CONSEQUENCE / LIMIT
- GLACIAL SEA-LEVEL FALL → exposed platform
- WAVE PLANATION → foundation shaping

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → combine sea level, biology and tectonics.

05

STAGE 05 BLEACHING OUTCOME FORK

BLEACHING OUTCOME FORK · POLLUTION STRESS · SYMBIONT LOSS · PALE OR WHITE COLONY · STRESS EASES · POSSIBLE RECOVERY · STRESS PERSISTS · DISEASE, MORTALITY, EROSION

THERMAL / LIGHT / POLLUTION STRESS

SYMBIONT LOSS → pale or white colony → STRESS EASES → possible recovery → STRESS PERSISTS → disease, mortality, erosion

CORE CLAIM
- THERMAL / LIGHT / POLLUTION STRESS
- SYMBIONT LOSS → pale or white colony

MECHANISM
- STRESS EASES → possible recovery
- STRESS PERSISTS → disease, mortality, erosion

CONSEQUENCE / LIMIT
- THERMAL / LIGHT / POLLUTION STRESS
- SYMBIONT LOSS → pale or white colony

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STRESS EASES → possible recovery; STRESS PERSISTS → disease, mortality, erosion.

06

STAGE 06 REEF-SERVICE CHAIN

REEF-SERVICE CHAIN · REEF COMPLEXITY · HABITAT + FISHERIES · REEF CREST · WAVE-ENERGY ATTENUATION · CARBONATE PRODUCTION · ISLAND SEDIMENT · LIVELIHOOD + EXPOSURE RISK

REEF COMPLEXITY → habitat + fisheries → REEF CREST → wave-energy attenuation → CARBONATE PRODUCTION → island sediment → DEGRADATION → livelihood + exposure risk

CORE CLAIM
- REEF COMPLEXITY → habitat + fisheries
- REEF CREST → wave-energy attenuation

MECHANISM
- CARBONATE PRODUCTION → island sediment
- DEGRADATION → livelihood + exposure risk

CONSEQUENCE / LIMIT
- REEF COMPLEXITY → habitat + fisheries
- REEF CREST → wave-energy attenuation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CARBONATE PRODUCTION → island sediment; DEGRADATION → livelihood + exposure risk.

07

STAGE 07 INDIA ISLAND CONTRAST

INDIA ISLAND CONTRAST · TECTONIC ARC, HIGH RELIEF, SEISMIC SETTING · LOW CORAL ATOLLS AND REEFS · TEN DEGREE · NICOBAR DIVIDE · NINE DEGREE · MAIN LAKSHADWEEP DIVIDE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
A&N → tectonic arc, high relief, seismic setting	TEN DEGREE → Andaman / Nicobar divide	A&N → tectonic arc, high relief, seismic setting
LAKSHADWEEP → low coral atolls and reefs	NINE DEGREE → Minicoy / main Lakshadweep divide	LAKSHADWEEP → low coral atolls and reefs

CORE CLAIM
- A&N → tectonic arc, high relief, seismic setting
- LAKSHADWEEP → low coral atolls and reefs

MECHANISM
- TEN DEGREE → Andaman / Nicobar divide
- NINE DEGREE → Minicoy / main Lakshadweep divide

CONSEQUENCE / LIMIT
- A&N → tectonic arc, high relief, seismic setting
- LAKSHADWEEP → low coral atolls and reefs

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NINE DEGREE → Minicoy / main Lakshadweep divide.

08

STAGE 08 GREAT NICOBAR VALUE MAP

GREAT NICOBAR VALUE MAP · TECTONIC ISLAND · EARTHQUAKE-TSUNAMI EXPOSURE · BIOSPHERE RESERVE · CORE-BUFFER-TRANSITION LOGIC · REEFS + BEACHES + FORESTS · CONNECTED HABITATS · INDIGENOUS COMMUNITIES

TECTONIC ISLAND → earthquake-tsunami exposure

BIOSPHERE RESERVE → core-buffer-transition logic → REEFS + BEACHES + FORESTS → connected habitats → INDIGENOUS COMMUNITIES → rights and livelihood

CORE CLAIM
- TECTONIC ISLAND → earthquake-tsunami exposure
- BIOSPHERE RESERVE → core-buffer-transition logic

MECHANISM
- REEFS + BEACHES + FORESTS → connected habitats
- INDIGENOUS COMMUNITIES → rights and livelihood

CONSEQUENCE / LIMIT
- TECTONIC ISLAND → earthquake-tsunami exposure
- BIOSPHERE RESERVE → core-buffer-transition logic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REEFS + BEACHES + FORESTS → connected habitats.

09

STAGE 09 PROJECT-STATUS FIREWALL

PROJECT-STATUS FIREWALL · PORT + AIRPORT + TOWNSHIP + POWER · CLEARANCE CONDITIONS · SAFEGUARDS TO IMPLEMENT · 12 FEB 2026 · FINAL FOREST APPROVAL NOT GRANTED · OFFICIAL CLAIM IS NOT MEASURED OUTCOME · CLOSE DISTINCTION

PROPOSAL → port + airport + township + power

CLEARANCE CONDITIONS → safeguards to implement → 12 FEB 2026 → final forest approval not granted → RULE → official claim is not measured outcome

CORE CLAIM
- PROPOSAL → port + airport + township + power
- CLEARANCE CONDITIONS → safeguards to implement

MECHANISM
- 12 FEB 2026 → final forest approval not granted
- RULE → official claim is not measured outcome

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Coral island is not coral reef, atoll is not every ring-shaped island, and bleaching is stress-driven
- Great Nicobar project components, tribal reserve, biosphere and statutory clearance statuses remain separate.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → official claim is not measured outcome.

10

STAGE 10 ISLAND APPRAISAL GATE

ISLAND APPRAISAL GATE · STRATEGIC AND ECONOMIC RATIONALE · HAZARD, FRESHWATER AND CARRYING CAPACITY · RIGHTS + CUMULATIVE EFFECTS · ALTERNATIVES, SAFEGUARDS, MONITORING · EVIDENCE LIMIT · DARWINIAN SUBSIDENCE IS A MODEL, NOT A UNIVERSAL SINGLE

NEED → strategic and economic rationale

PLACE → hazard, freshwater and carrying capacity → PEOPLE / ECOLOGY → rights + cumulative effects → DECISION → alternatives, safeguards, monitoring

CORE CLAIM
- NEED → strategic and economic rationale
- PLACE → hazard, freshwater and carrying capacity

MECHANISM
- PEOPLE / ECOLOGY → rights + cumulative effects
- DECISION → alternatives, safeguards, monitoring

CONSEQUENCE / LIMIT
- EVIDENCE LIMIT: Darwinian subsidence is a model, not a universal single pathway
- reef condition and project impacts need site/date/source evidence, cumulative-risk analysis and explicit uncertainty rather than benefit-versus-environment binaries.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Darwinian subsidence is a model, not a universal single pathway; reef condition and project impacts need site/date/source evidence, cumulative-risk analysis and explicit uncertainty rather than benefit-versus-environment binaries.

11

STAGE 11 ISLAND ANSWER SPINE

ISLAND ANSWER SPINE · ORIGIN AND REEF TYPE · ARC, ATOLL, CHANNEL AND LAGOON · HEAT, HAZARD, RIGHTS AND STATUS · ISLAND-SPECIFIC AND PRECAUTIONARY

CLASSIFY → origin and reef type → MAP → arc, atoll, channel and lagoon → ASSESS → heat, hazard, rights and status → VERDICT → island-specific and precautionary

CORE CLAIM
- CLASSIFY → origin and reef type
- MAP → arc, atoll, channel and lagoon

MECHANISM
- ASSESS → heat, hazard, rights and status
- VERDICT → island-specific and precautionary

CONSEQUENCE / LIMIT
- CLASSIFY → origin and reef type
- MAP → arc, atoll, channel and lagoon

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ASSESS → heat, hazard, rights and status; VERDICT → island-specific and precautionary.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · ARAKAN YOMA EXTENSION: TECTONIC/FOLD-BELT LINK. · NICOBAR ISLANDS BELONG TO THE TECTONIC ISLAND ARC AND · BARREN ISLAND · INDIA'S ONLY ACTIVE VOLCANO. · VOLCANIC ISLAND; BARREN ISLAND IS INDIA'S ONLY HISTORICALLY ACTIVE · ENTIRELY CORAL ATOLLS IN ARABIAN SEA. · A&N REGION LIES IN A MAJOR EARTHQUAKE ZONE.

OPTIONAL SOURCE DEPTH
- Andaman = Arakan Yoma extension; tectonic/fold-belt link.
- Nicobar islands belong to the tectonic island arc and are fringed by coral reefs; they are not simply coral atolls.
- Barren Island = India's only active volcano.
- Narcondam = volcanic island; Barren Island is India's only historically active volcano.

ADVANCED DEBATE
- Lakshadweep = entirely coral atolls in Arabian Sea.
- A&N region lies in a major earthquake zone.
- Saddle Peak is in North Andaman and is the archipelago's highest point.
- NOT: Lakshadweep is volcanic like Barren Island → Lakshadweep is coral/atoll ; Barren Island is volcanic.

LEGACY / NUANCE
- NOT: Andaman Islands are coral like Lakshadweep → Andamans are linked to the Arakan Yoma fold chain .
- NOT: Narcondam is India's active volcano → Barren Island is active; Narcondam is dormant.
- NOT: Bay of Bengal and Arabian Sea islands have the same origin → Bay islands are tectonic/volcanic/coral mix; Lakshadweep is coral.
- NOT: Island geography is only physical → Indian islands also have strategic, ecological and disaster-risk dimensions.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels